Unit -3
system & Requirement engineering

Topics

- What is Computer Based System?
- Different System elements
- Requirements Engineering Process
- Business Process Engineering (BPE)
- System Engineering Hierarchy
- What is Business Process?
- What is Product Engineering?
- System Modelling
- What is Requirement Engineering?
- Requirement Engineering Tasks

What is Computer Based System?

- The **computer-based system** consists of all components necessary to
- capture, process, transfer, store, display, and manage information.

- Components include **software**, processors, networks, buses, firmware, application-specific integrated circuits, storage devices, and humans.

What is Computer Based System?

- we define a computer-based system as
- A set or arrangement of elements that are organized to accomplish some predefined goal by processing information.

There are different system elements

- **Software**
- **Hardware**
- **People**
- **Database**
- **Documentation**
- **Procedures**

System Elements

Software:

- is a set of programs, Computer programs, data structures, and related documentation.
- that serve to effect the logical method, procedure, or control that is required.

Hardware:

- Is a electronic devices that provide computing capability, the interconnectivity devices (e.g., network switches, telecommunications devices) that enable the flow of data, and
- electromechanical devices (e.g., sensors, motors, pumps) that provide external world function.

System Elements

People:

- Users and operators of hardware and software.

Database:

- A large, organized collection of information that is accessed via software.

Documentation:

- Descriptive information (e.g., hardcopy manuals, on-line help files, Web sites) that shows the use and/or operation of the system.

Procedures:

- The steps that define the specific use of each system element or the procedural context in which the system resides.

Requirements Engineering Process

- Requirement Engineering is the process of defining, documenting and maintaining the requirements.
- It is a process of gathering and defining service provided by the system.

Requirements Engineering Process consists of the following main activities:

- Requirements elicitation
- Requirements specification
- Requirements verification and validation
- Requirements management

Requirements Elicitation

- It is related to the various ways used to gain knowledge about the project domain and requirements.

- The various sources of domain knowledge include customers, business manuals, the existing software of same type, standards and other stakeholders of the project.
- The techniques used for requirements elicitation include interviews, brainstorming, task analysis, Delphi technique, prototyping, etc.
- domain knowledge of the analyst and thus helps in providing input to the next stage.

Requirements specification

- This activity is used to produce formal software requirement models.
- All the requirements including the functional as well as the non-functional requirements and the constraints are specified by these models in totality.
- During specification, more knowledge about the problem may be required which can again trigger the elicitation process.

- The models used at this stage include ER diagrams, data flow diagrams(DFDs), function decomposition diagrams(FDDs), data dictionaries, etc.

Requirements verification and validation

- **Verification:** It refers to the set of tasks that ensures that the software correctly implements a specific function.

- **Validation:** It refers to a different set of tasks that ensures that the software that has been built is traceable to customer requirements.

Requirements verification and validation

- If requirements are not validated, errors in the requirement definitions
- would propagate to the successive stages resulting in a lot of modification and rework.

The main steps for this process include:

- The requirements should be consistent with all the other requirements i.e. no two requirements should conflict with each other.
- The requirements should be complete in every sense.
- The requirements should be practically achievable.
- Reviews, buddy checks, making test cases, etc. are some of the methods used for this.

Requirements management

- Requirement management is the process of analyzing, documenting, tracking, prioritizing and agreeing on the requirement and controlling the communication to relevant stakeholders.

- This stage takes care of the changing nature of requirements.
- It should be ensured that the SRS is as modifiable as possible so as to incorporate changes in requirements specified by the end users at later stages too.

- This is important part of the requirements engineering process.

Business Process Engineering (BPE)

- Business process engineering is a way in which organizations study their current business processes and develop new methods to improve productivity, efficiency, and operational costs.

- Business process engineering focuses on new business processes, how to diagnose problems with an organization's current methodology,

- and how to redesign, reconstruct, and monitor processes to ensure they are effective.

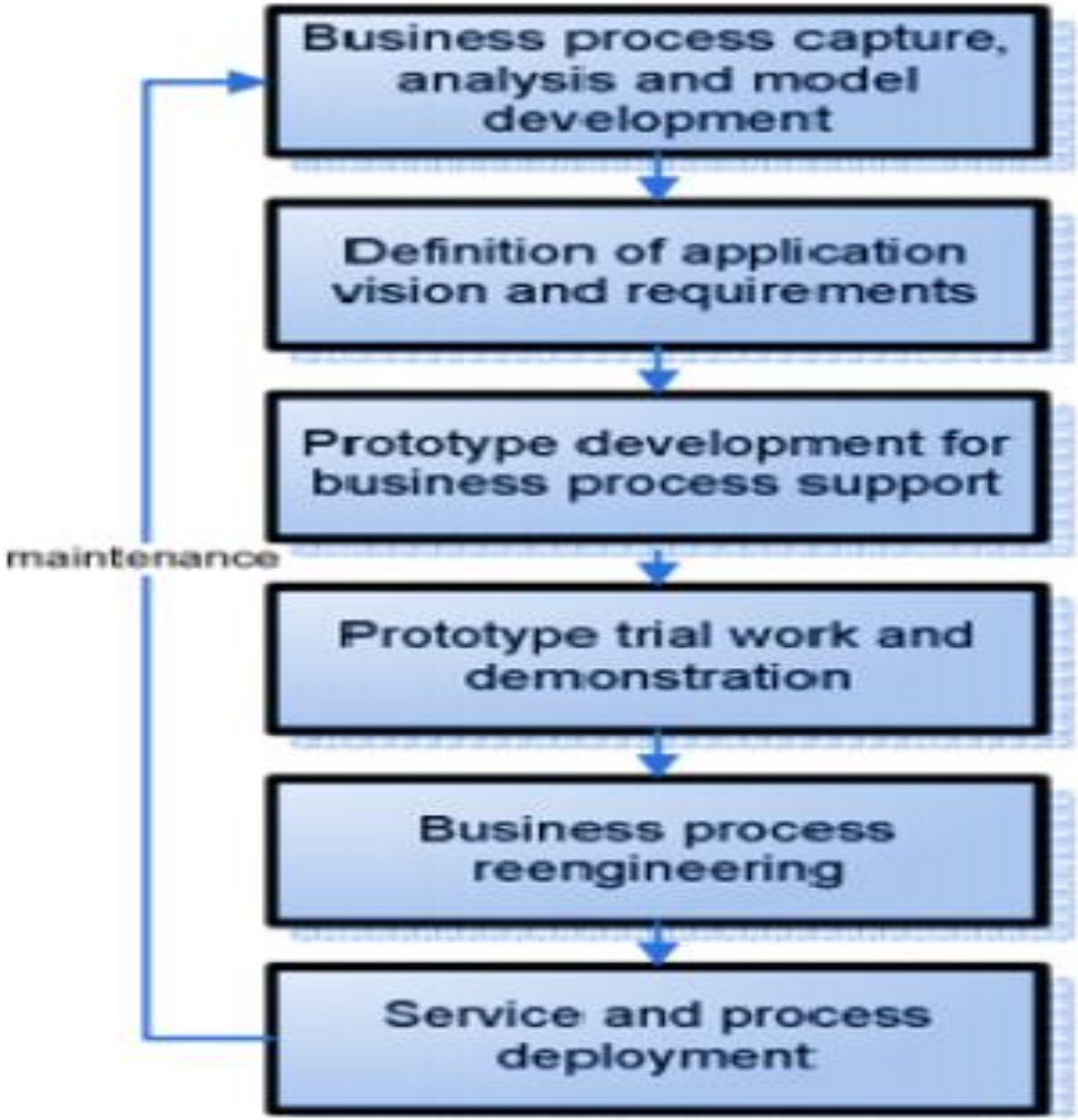
Business Process Engineering (BPE)

- Business Process Engineering (BPE) uses a proven systematic approach based on the latest experiences and research to achieve significant improvements.

- The BPE process helps clients fundamentally rethink and reinvent the business processes needed to achieve the firm's strategic objectives through the maximum use of enabling technologies and organizational strategies.

- A BPE effort can result in 15% to 50% improvement in performance of the targeted business processes, depending upon whether a reengineering or an improvement approach is used in the effort.

Business Process Engineering (BPE)



BPE Approach

- Business Process Engineering (BPE) Approach includes:

Understanding the Present Mode of Operation (PMO): to analyze your current processes, technologies and systems.

- The result will be creation of a detailed PMO business process model showing interrelationships and dependencies between people, systems, and processes.

Determining the Future Mode of Operation (FMO): We'll work with you to build an advisory team to define an FMO business process model based on your business objectives and our combined knowledge of industry best practices.

BPE Approach

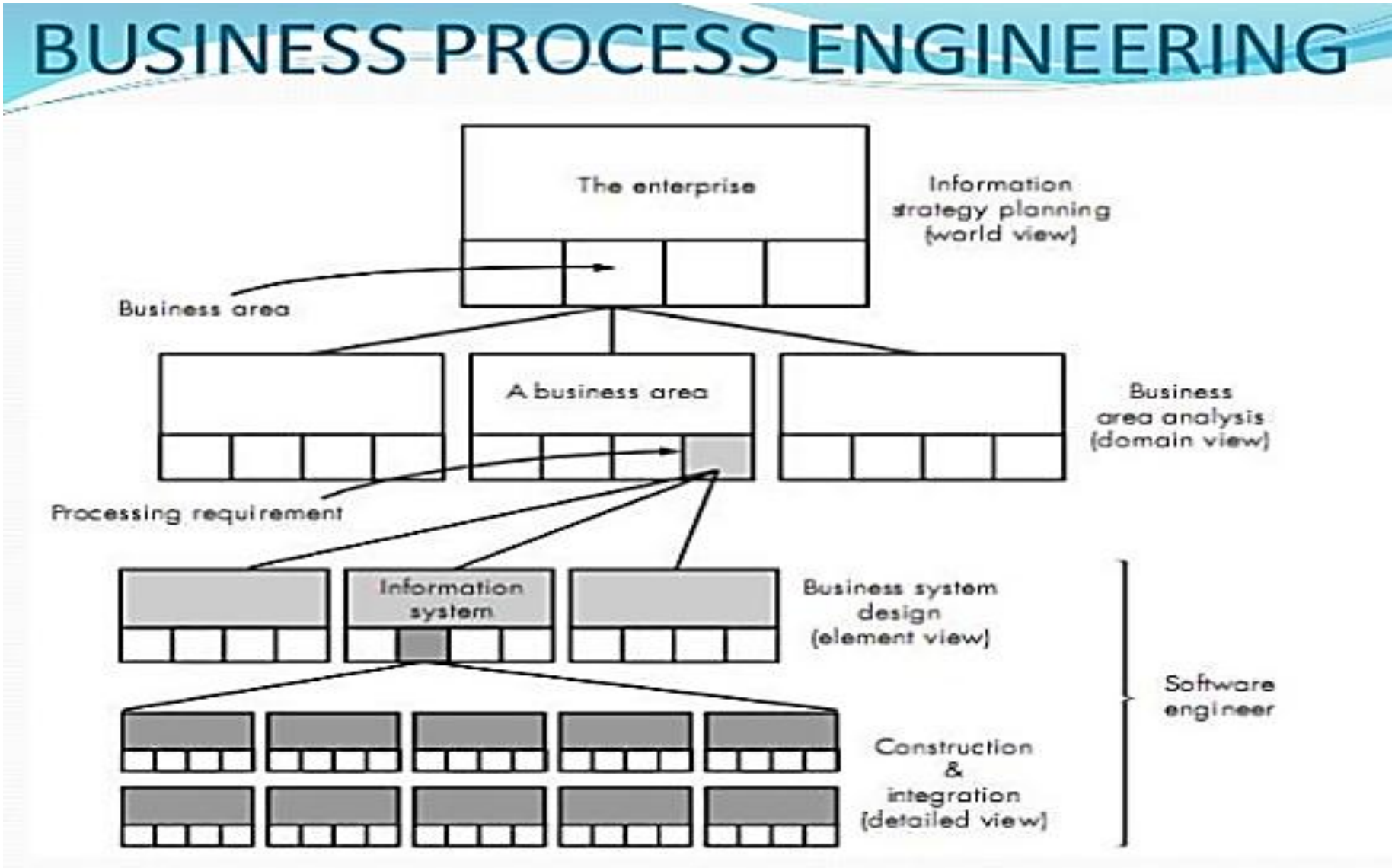
Gap Analysis and Transition Plan: A gap analysis of needed business process improvements and transition to the FMO plan will be developed.

- You'll gain an understanding of the business strategy, timing, personnel, and system/process evolution that will take place.

Implementation: By trialing and deploying new system or operational process improvements, we'll help you determine whether the intended results will be achieved.

- If additional system and operational process improvements need to be made, we'll repeat the appropriate PMO, FMO, Gap Analysis and Transition Planning steps as necessary.

Business Process Engineering example



System Engineering Hierarchy

The system engineering process usually begins with a world view.

It is a collection of top-down and bottom-up method to navigate the hierarchy.

There are different part of system engineering hierarchy.

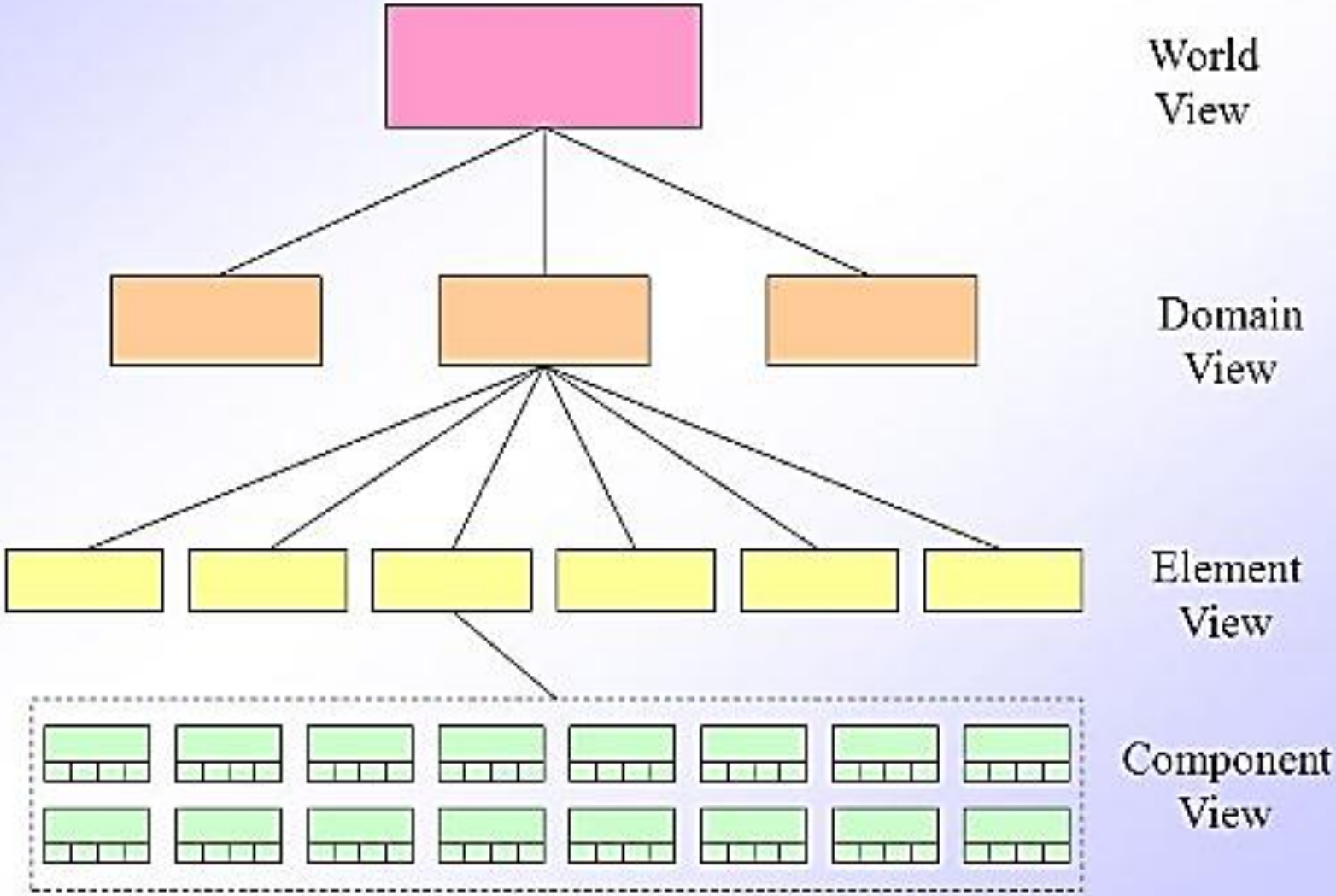
World view (WV) is a set of domain means system

Domain view (Di) like hardware, software, people etc.

Element view (Ej) like data, function, behavior etc.

Detailed view like integrate with system etc.

System Engineering Hierarchy



System Engineering Hierarchy

The world view (WV) is composed of a set of domains (Di), which can each be a system or system of systems in its own right.

$$WV = \{D_1, D_2, D_3, \ldots, D_n\}$$

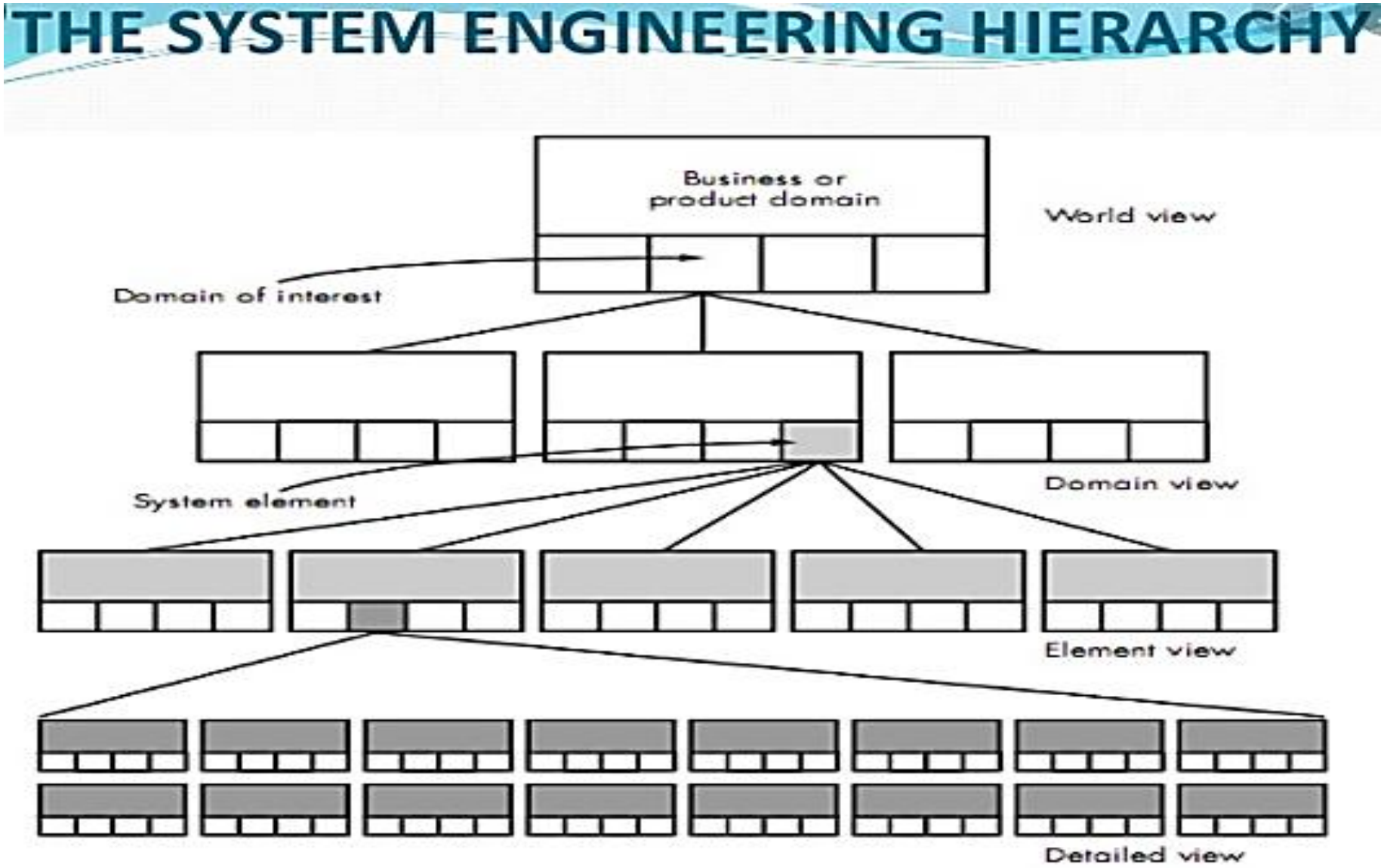
Each domain is composed of specific elements (Ej) each of which serves some role in accomplishing the objective and goals of the domain or component:

$$D_i = \{E_1, E_2, E_3, \ldots, E_m\}$$

Finally, each element is implemented by specifying the technical components (Ck) that achieve the necessary function for an element:

$$E_j = \{C_1, C_2, C_3, \ldots, C_k\}$$

Final figure



What is Business process?

- Business process engineering: The system engineering process is called business process engineering when the context of the engineering work focuses on a business enterprise.

What is Product engineering?

- Product engineering: When a product is to be built, the process is called product engineering.

- Both business process engineering and product engineering attempt to bring order to the development of computer-based systems.

Product engineering

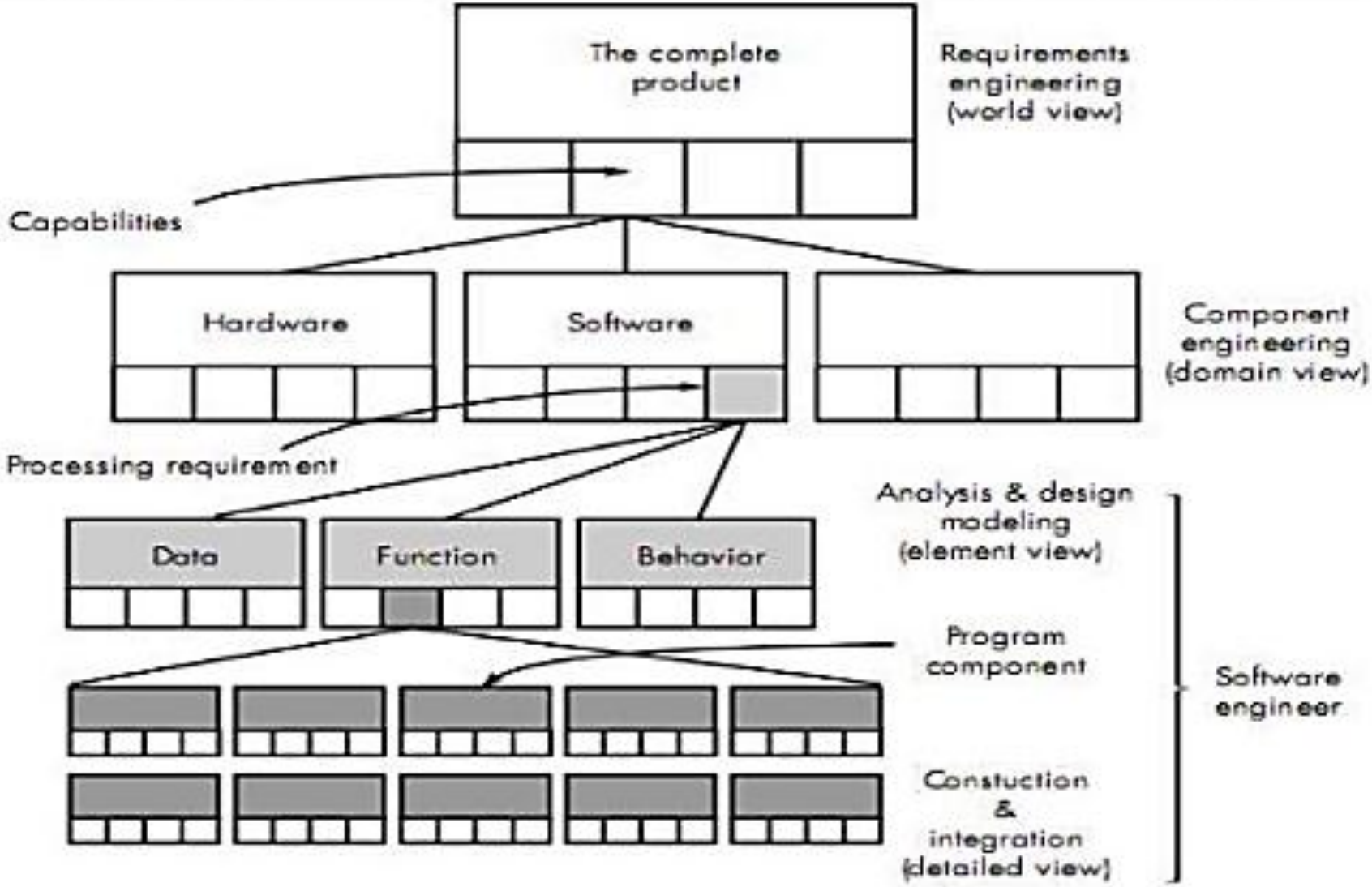
The goal of product engineering is to translate the customer's desire for a set of defined capabilities into a working product.

To achieve this goal, product engineering—like business process engineering—must derive architecture and infrastructure.

- **The architecture** encompasses four distinct system components: software, hardware, data (and databases), and people.

- **The infrastructure** includes the technology required to tie the components together and the information that is used to support the components.

The product engineering hierarchy



What is System Modeling?

- **System Modeling** is the process of developing abstract models of a system with model presenting a different view or perspective of that system.

OR

- **A System Model** represent aspects of a system and its environment.

OR

- **System Modeling** is a mean of representing a world view to detailed view of the system using same kind of Graphical Notation.

Features of a System Modeling

- define the processes that serve the needs of the view under consideration
- represent the behavior of the processes and the assumptions on which the behavior is based
- explicitly define both a internal and external input to the model
- represent all linkages(input/output) that will enable engineer to better understand the view.

System Modeling

- To construct a model, the engineers should consider a number of restraining factors

- - assumptions
- - simplifications
- - limitations
- - constraints
- - preferences

System Modeling

- **Assumptions:**
  It enables a model to reflect the problem in a reasonable manner by reducing the number of possible **permutations and variations.**

  - **Example:** Representation of 3D human forms

    In this input domain maybe that the system engineer makes certain assumptions about the range of allowable human movements.
  - **(leg cannot be wrapped around torso) so that range of Input and processing can be Limited**.

System Modeling

- **Simplifications:**
  That enables the model to be created in a timely manner.

  - **Example:** A System Engineer is modeling the needs of the service organizational and is working to understand the flow of information that issues a service order.

  - Although a service order can be derived from many origins, the engineer categorizes only two sources.

  - **To accept Internal Demand and External Request** This enables a simplified partitioning of input that is required to generate the service order.

System Modeling

- **Limitations:** That help to bound the system.
  - **Example:** An aircraft avionics system is being modeled for future aircraft.
  - Is the aircraft will be a two-engine design, the monitoring domain for propulsion will be modeled to accommodate a maximum of two engines and associative terminated system.

- **Constraints:** That will guide the manner in which the model is created and the approach taken when the model is implemented.

- **Preferences:** It indicates the preferred architecture for all data, functions and technology.

What is Requirement Engineering?

- The requirements themselves are the description of the system services and constraints that are generated during the requirement engineering process.

- The process of establishing the services that the customer requires from a system and the constraints under which it operates and is developed.

What is Constraints?

- Constraints are things that we live with in our daily lives, accept them, and move on.

- Constraints are things that affect your product but are "**out of your control**".

- Careful **planning**, collaboration with internal and external stakeholders, and holding yourself responsible for what you say and do is the professional way of doing things.

- For example Economical Constraints, Managerial Constraints, Technical Constraints, Social Constraints.

Types of requirement

User requirements –

- Statements in natural language plus diagrams of the services the system provides and its operational constraints. Written for customers.
  - For example user interface

System requirements –

- A structured document setting out detailed descriptions of the system's functions, services and operational constraints.
- Defines what should be implemented so may be part of a contract between client and contractor.
  - For example generate sales report, customer data etc.

Functional and non-functional requirements

- **Functional requirements –** Statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations.

- **Non-functional requirements –** Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.

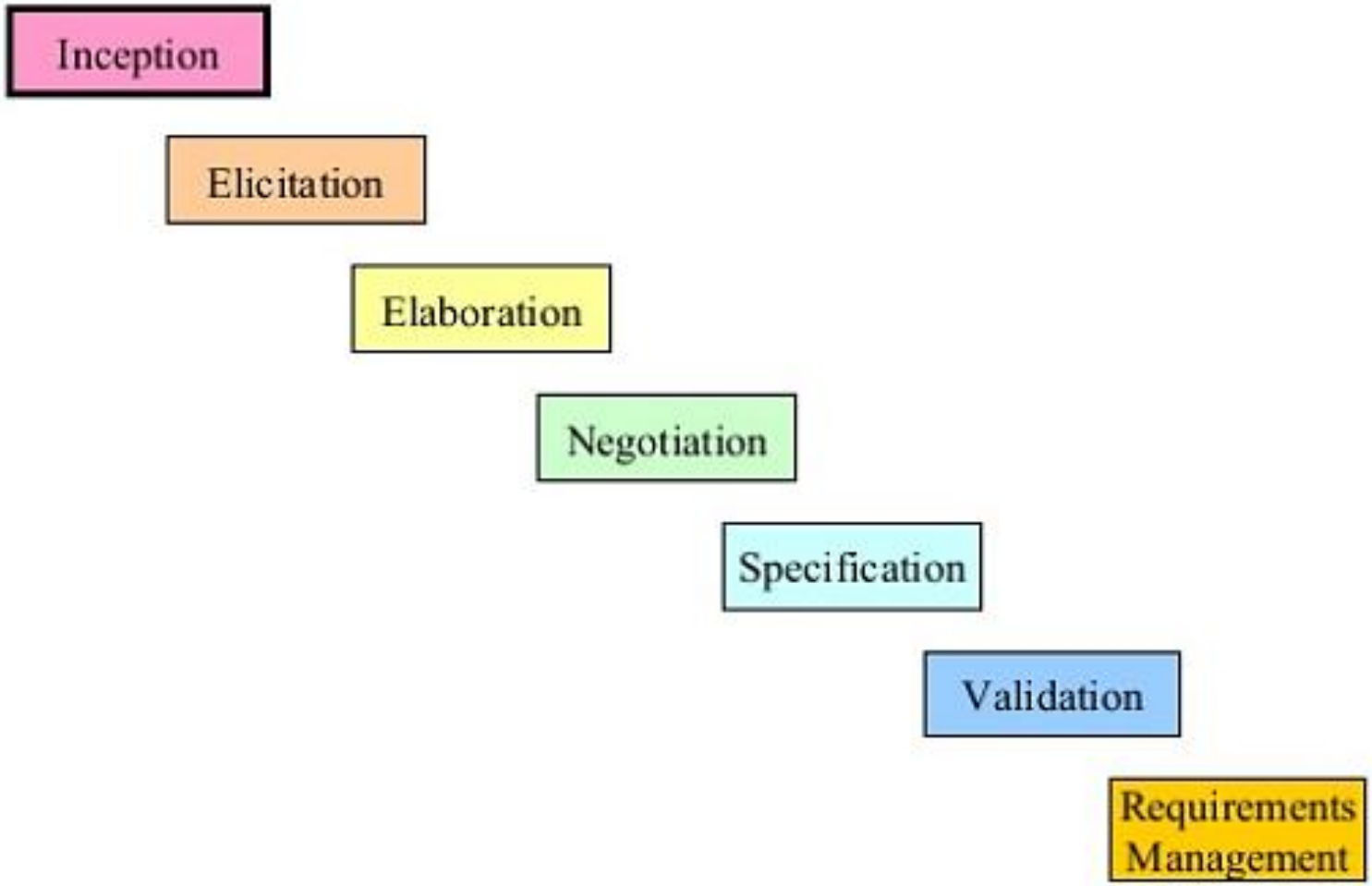
- **Domain requirements –** Constraints on the system from the domain of operation

Requirements Engineering

Requirements engineering tasks:

- Inception
- Elicitation
- Elaboration
- Negotiation
- Specification
- Validation
- Requirements Management

Requirement Engineering Task



Inception Task

- During inception, the requirements engineer asks a set of questions to establish…
  - A **basic understanding of the problem**
  - The people **who want** a solution
  - The **nature of the solution** that is desired
  - The effectiveness of preliminary communication and collaboration between the **customer and the developer**
- Through these questions, the requirements engineer needs to…
  - Identify the **stakeholders**
  - Recognize **multiple viewpoints**
  - Work toward **collaboration**(give and take policy)
  - Initiate the **communication**

Elicitation Work Products

The work products will vary depending on the system, but should include one or more of the following items

- A statement of **need and feasibility**
- A bounded statement of **scope** for the system or product
- A list of customers, users, and other **stakeholders** who participated in requirements elicitation
- A description of the **system's technical environment**
- A **list of requirements** (organized by function) and the domain constraints that apply to each
- A set of preliminary **usage scenarios** (in the form of use cases) that provide insight into the use of the system or product under different operating conditions
- Any **prototypes** developed to better define requirements

Elicitation Task

- Eliciting requirements is difficult because of
  - **Problems of scope** in **identifying the boundaries** of the system or specifying too much technical detail rather than overall system objectives
  - **Problems of understanding** what is wanted, what the problem domain is, and what the computing environment can handle (Information that is believed to be **"obvious" is often omitted)**
  - **Problems of volatility** because the requirements change over time
- Elicitation may be accomplished through two activities
  - Collaborative **requirements gathering**
  - Quality **function deployment**

Elaboration Task

- During elaboration, the software engineer takes the information obtained during **inception and elicitation** and begins to **expand and refine it**
- Elaboration focuses on developing a **refined technical** model of software functions, features, and **constraints**
- It is an analysis modeling task
  - **Use cases** are developed
  - **Domain classes** are identified along with their **attributes** and **relationships**
  - State **machine diagrams** are used to capture the life on an object
- The end result is an **analysis model** that defines the functional, informational, and behavioral model of the problem

Negotiation Task

- During negotiation, the software engineer **reconciles the conflicts** between **what the customer wants** and **what can be achieved** given limited business resources
- Requirements are ranked (**i.e., prioritized**) by the customers, users, and other stakeholders
- **Risks associated** with each requirement are identified and analyzed
- Rough guesses of **development effort** are made and used to assess the impact of each requirement on project cost and **delivery time**
- Using an iterative approach, requirements are eliminated, combined and/or modified so that each party achieves some **measure of satisfaction**

Specification Task

- A specification is the **final work product** produced by the requirements engineer
- It is normally in the form of a **software requirements specification**
- It serves as the **foundation** for subsequent software engineering activities
- It describes the **function and performance** of a computer-based system and the constraints that will govern its development
- It **formalizes requirements** of the proposed software in both a graphical and textual format

Validation Task

- During validation, the work products produced as a result of requirements engineering are assessed for quality
- The specification is examined to ensure that
  - all software requirements have been **stated unambiguously**
  - inconsistencies, omissions, and errors have been detected and corrected
  - the work products conform to the standards established for the process, the project, and the product
- The formal technical review serves as the primary requirements validation mechanism
  - Members include software engineers, customers, users, and other stakeholders

Requirements Management Task

- During requirements management, the project team performs a set of activities to i**dentify, control, and track requirements** and changes to the requirements at any time as the project proceeds
- Each requirement is assigned a **unique identifier**
- The requirements are then placed into one or more **traceability tables**
- These tables may be stored in a database that relate features, sources, dependencies, subsystems, and interfaces to the requirements
- A **requirements traceability table** is also placed at the end of the software requirements specification

Important Questions

- What is Computer Based System? Define different System elements.
- Write a short note on Requirements Engineering Process.
- Explain Business Process Engineering (BPE).
- Discuss System Engineering Hierarchy.
- Define Terms: Business process, Product engineering, Requirement engineering, System modelling etc.
- Write a short note on System modelling
- What is Requirement engineering? Explain Requirement engineering tasks.

- If any query free feel to ask